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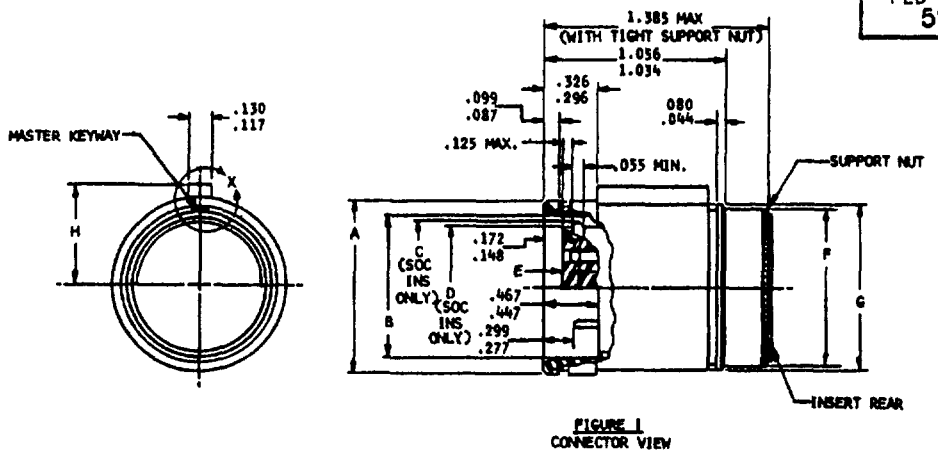
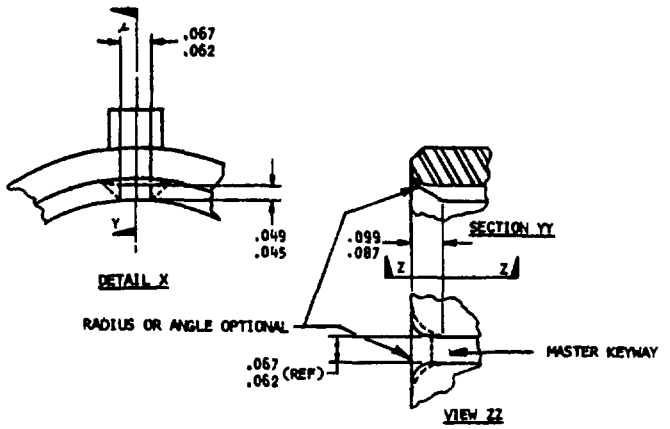
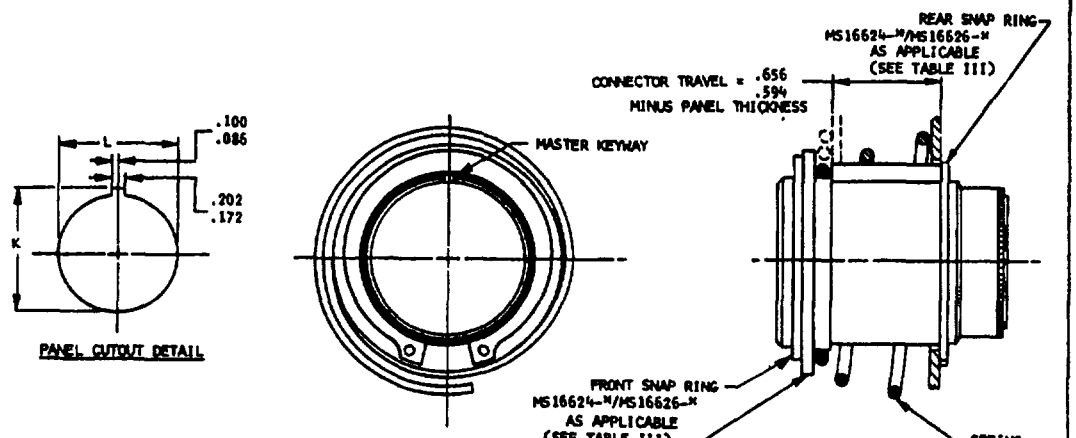


FIGURE I
CONNECTOR VIEW





PANEL CUTOUT DETAIL

FIGURE II
CONNECTOR ASSEMBLY
VIEW

EXAMPLE OF PART NO. MS3445E 19 - 50 S Y

MS NUMBER	CLASS	SIZE	INSERT ARRANGEMENT	CONTACT STYLE	ALTERNATE INSERT POSITION

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P A NAVY-AS Other Cust	International Interest	TITLE CONNECTOR, PLUG, ELECTRIC, CRIMP TYPE, RACK AND PANEL, SERIES 2 (CLASS E)	MILITARY STANDARD MS 3445(NAVY)
Procurement Specification MIL-C-81703		SUPERSEDES:	SHEET 1 OF 5

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TABLE I
CONNECTOR AND PANEL
CUTOUT DIMENSIONS

SHELL SIZE DASH NO	A DIA. OVER FRONT OF SHELL	B I D SHELL	C DIA INSERT SEAL (SOC INS ONLY)	D DIA OF INSERT (SOC INS ONLY)	E POINT OF SPRING CONTACT	F MAX DIA REAR	G DIA OVER REAR OF SHELL	H DIM FROM ℄ TO TOP OF KEY
-3	641/ 628	454/ 446	380/ .362	354/ 334	.138 MAX MEASURE WITH SQUARE NOSE PIN HAVING BASIC DIAMETER OF MATING CONTACT	625	615/ 595	442/ 422
-7	797/ 784	589/ 581	496/ .478	470/ 450		702	760/ 740	504/ 484
-12	920/ 907	723/ 715	.636/ .618	610/ 590		828	867/ 847	567/ 547
-19	1.048/1 035	862/ 854	776/ .756	750/ 726		953	1 010/ 990	639/ 619
-27	1 189/1 176	1 017/1 009	931/ 911	905/ .880		1 102	1 150/1 130	709/ 689
-37	1 356/1 343	1 139/1 131	1 053/ 1.038	1 027/ 1.003		1 266	1 322/1 302	796/ 776

SHELL SIZE DASH NO	K	L
-3	858/ 828	.686/.656
-7	999/ 969	.827/ 797
-12	1.124/1 094	.968/.938
-19	1 249/1 219	1.077/1.047
-27	1 390/1 360	1.218/1 188
-37	1 546/1 516	1.390/1.360

SHELL SIZE DASH NO	* WEIGHT MAX (LBS)			
	CLASS E		CLASS F	
	PIN INSERT	SOC INSERT	PIN INSERT	SOC INSERT
-3				
-7				
-12	WEIGHTS WILL BE SUPPLIED IN LATER REVISION			
-19				
-27				
-37				

*WEIGHT LISTED IS FOR CONNECTOR WITH SNAP RINGS, BACKING WASHER, SPRING, AND MAX DENSITY INSERT ARRANGEMENT LESS CONTACTS.

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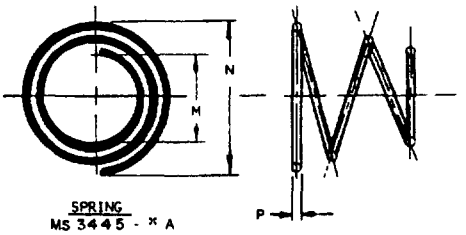
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Procurement Specification MIL-C-81703		SUPERSEDES:	SHEET 2

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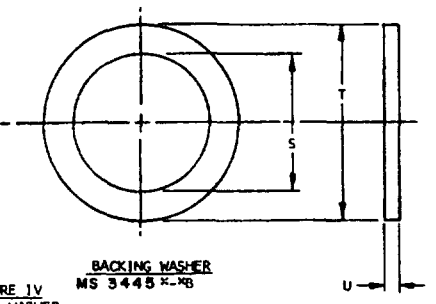
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SPRING
MS 3445 - * A

FIGURE III
SPRING DETAIL



BACKING WASHER
MS 3445 *-*B

FIGURE IV
BACKING WASHER

SPRING MATERIAL STEEL
FINISH BLACK OXIDE PER MIL-C-13924, CLASS 1
CONSTRUCTION SPRING MUST BE MADE SO COILS WILL COLLAPSE WITHIN EACH OTHER AT A SOLID HEIGHT EQUAL TO THE WIRE DIAMETER

TABLE II
SPRING AND BACKING WASHER DIMENSIONS

SHELL SIZE DASH NO	A -- SPRING								B — BACKING WASHER		
	M MIN	N MAX	P	APPROX FREE LENGTH	TOTAL TURNS (REF)	SPG FORCE AT "R" HEIGHT (LBS MIN)	R ± 0'0	SPG FORCE AT P + 020 HEIGHT (LBS MAX)	S	T	U
-3	645	1 220	071/ 059	800	3	8	220	18	660/ 649	.891/ 869	070/ 050
-7	775	1 350	073/ 061	1 250	3	10	230	18	815/ 804	1.172/1 150	110/ 090
-12	905	1 565	097/ 085	938	2 1/3	21	250	30	932/ 921	1.263/1.241	091/ 071
-19	1 025	1 705	097/ 085	1 875	2 1/2	27	250	45	1 060/1 049	1 391/1 369	081/ 061
-27	1 175	1 855	101/ 089	1 625	2 5/8	28	260	46	1 198/1 190	1 529/1.507	072/ 052
-37	1 345	2 105	126/ 114	1 250	2 1/2	28	280	48	1 367/1 357	1 816/1.794	060/ 040
-61	1 675	2 835	141/ 129	2.250	2 1/2	54	295	77	1 685/1.674	2.150/2.118	.072/.052

EXAMPLE OF PART NO MS 3445- 3 A
MS NUMBER — SIZE — SPRING

EXAMPLE OF PART NO. MS 3445 * - 3 B
MS NUMBER — CLASS — SIZE — WASHER
E= CAD PLATE
L= NICKEL PLATE

TABLE III
SNAP RING
PART NUMBERS

SHELL SIZE DASH NO	FRONT SNAP RING	REAR SNAP RING	
-3	MS16624-4062	MS16626-4059	EXCEPT WITH BLACK OXIDE FINISH PER MIL-C-13924, CLASS 1
-7	MS16624-4078	MS16624-4075	EXCEPT WITH BLACK OXIDE FINISH PER MIL-C-13924, CLASS 1
-12	MS16626-4093	MS16624-4087	EXCEPT WITH BLACK OXIDE FINISH PER MIL-C-13924, CLASS 1
-19	MS16624-4106	MS16624-4100	EXCEPT WITH BLACK OXIDE FINISH PER MIL-C-13924, CLASS 1
-27	MS16624-4118	MS16624-4112	EXCEPT WITH BLACK OXIDE FINISH PER MIL-C-13924, CLASS 1
-37	MS16624-4137	MS16624-4131	EXCEPT WITH BLACK OXIDE FINISH PER MIL-C-13924, CLASS 1
-61	MS16624-4168	MS16624-4162	EXCEPT WITH BLACK OXIDE FINISH PER MIL-C-13924, CLASS 1

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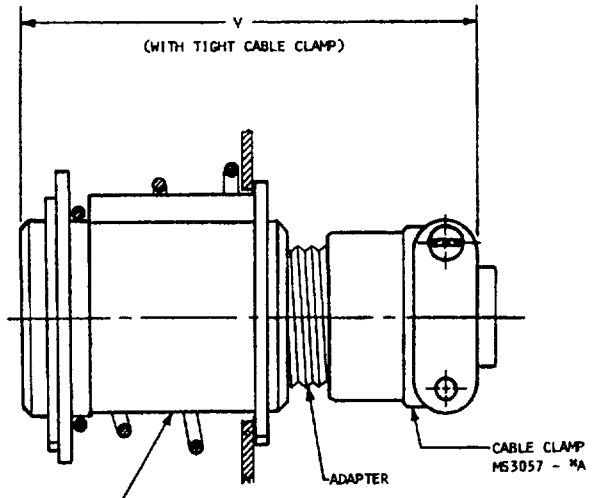


FIGURE V
CONNECTOR WITH ADAPTER
AND CABLE CLAMP

TABLE IV
ADAPTER AND CABLE
CLAMP INFORMATION

SHELL SIZE DASH NO.	V LENGTH MAX	ADAPTER PART NUMBER	CABLE CLAMP PART NUMBER
-3	2 234	MS3445-3C	MS3057-3A
-7	2 234	MS3445-7C	MS3057-4A
-12	2 297	MS3445-12C	MS3057-6A
-19	2 359	MS3445-19C	MS3057-8A
-27	2 359	MS3445-27C	MS3057-10A
-37	2 359	MS3445-37C	MS3057-12A

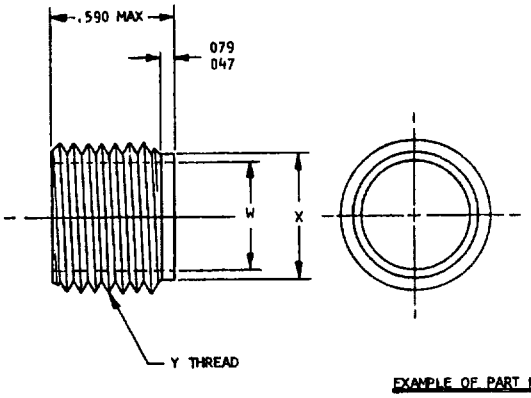


FIGURE VI
ADAPTER DETAIL

TABLE V
ADAPTER
DIMENSIONS

SHELL SIZE DASH NO	C -- ADAPTER		
	W I D	X	Y THREAD
-3	401/ 389	451/ 444	1/2-28 UNEF-2A
-7	506/ 494	572/ 565	5/8-24 UNEF-2A
-12	.651/ 639	688/ 681	3/4-20 UNEF-2A
-19	.781/ 769	813/ 806	7/8-20 UNEF-2A
-27	888/ 882	938/ 931	1-20 UNEF-2A
-37	1 052/1 040	1 118/1 111	1 3/16-18 UNEF-2A

EXAMPLE OF PART NO. MS 3445 - 12 C
MS NUMBER, SIZE, ADAPTER

THIS ADAPTER IS INTENDED FOR USE WITH CLASS E CONNECTORS, WITH CABLE CLAMP.

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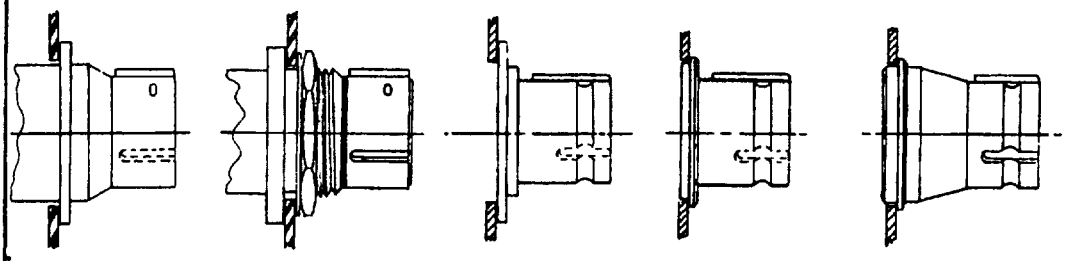
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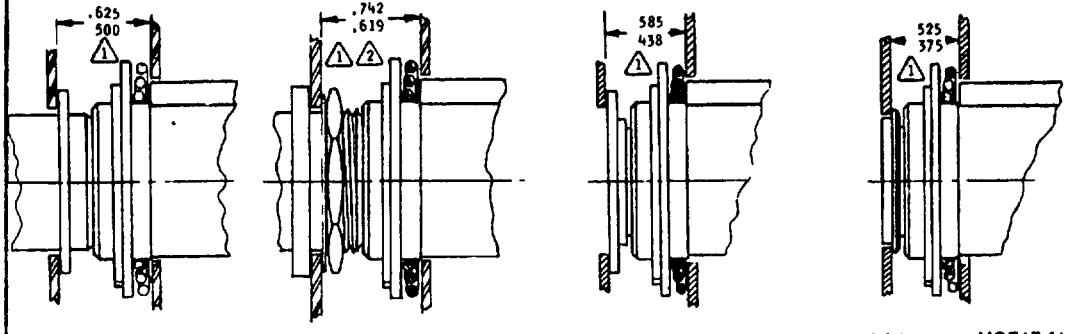
MS3130
MS3140
MS3424

MS3464

MS3132H
MS3466H

MS3134H
MS3463H

MS3145
MS3143



MS3445 MS3446 OR MATED WITH MS3130 MS3140 MS3424

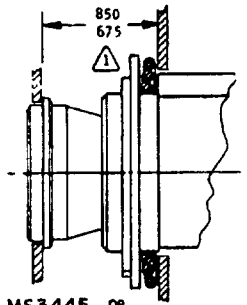
MS3445 MS3446 OR MATED WITH MS3464

MS3445 MS3446 OR MATED WITH MS3132H MS3466H

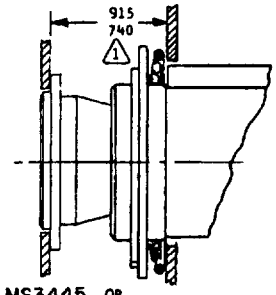
MS3445 MS3446 OR MATED WITH MS3134H MS3463H

(1) PANEL SEPARATION FOR DESIGN PURPOSES, IN HIGH VIBRATION AND/OR OTHER ADVERSE ENVIRONMENT APPLICATIONS, PANEL SEPARATION SHALL BE AS CLOSE TO THE MINIMUM ALLOWABLE DIMENSION AS FEASIBLE
(MAX PANEL DEFLECTION .016 INCLUDED IN THIS DIMENSION)

(2) DIMENSION SHOWN IS TO BACK OF PANEL FOR ACTUAL PANEL SEPARATION, THE PANEL THICKNESS ON RECEPTACLE SIDE MUST BE SUBTRACTED FROM THE DIMENSION SHOWN.

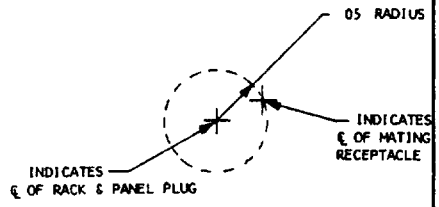


MS3445 MS3446 OR MATED WITH MS3143



MS3445 MS3446 OR MATED WITH MS3145

FIGURE VII
MATING INFORMATION



RACK & PANEL INSTALLATION SHALL BE DESIGNED TO MAINTAIN CONNECTOR ALIGNMENT WITHIN .05 IN. AS SHOWN ABOVE.

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